

Anxiety Disorders in Children: A Multiple-Baseline Evaluation of a Cognitive-Behavioral Treatment

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Four children diagnosed with Overanxious Disorder were treated individually with a 16-20 session, cognitive-behavioral treatment that was evaluated using assessments from multiple sources and a multiple-baseline across cases design for the self-reports. The children, aged 9-13 years, came from middle class backgrounds, and were within the limits of normal intelligence. Parent report, independent clinician's ratings, and child self-report were used to assess outcome. All children showed improvement on parent and independent clinician's ratings, as well as on self-report. Treatment-produced gains, as measured with parent and child report were, in general, maintained at 3-6 month follow-up. These outcomes suggest that a cognitive-behavioral approach to treating anxiety in children is useful, though a controlled comparison evaluation is needed.

There has been a growing concern accompanied by mounting evidence suggesting the need for interventions designed specifically for children. Indeed significant contributions have been made in terms of treatment strategies for the externalizing, aggressive, non-compliant, hyperactive, and impulsive child (e.g. Forehand & McMahon, 1981; Kendall & Braswell, 1985). However, treatment strategies and interventions for the internalizing, anxious, isolated, or withdrawn child have not been as fully developed. Kendall, Howard, and Epps (1988) noted the paucity of group comparison studies for childhood anxiety disorders.

Certainly children displaying significant levels of internalizing behaviors constitute a noticeable proportion of any child clinical population. Specifically, Morris and Kratochwill (1985) noted variable prevalence rates in various studies

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* A copy of the cognitive-behavioral treatment manual for anxious children is available from the second author.

* Only trait anxiety scores are reported for each child since the validity of the state-trait distinction has been called into question (Johnson & Melamed, 1979) and the state scale may be subject to the influence of specific events at the moment the child completes the measure.

of the incidence of clinical fears (<8%). Social withdrawal as evidenced by anxiety, isolation, hypersensitivity, depression and self-consciousness is seen in 10–20% of school-age children (Werry & Quay 1971). Recently, Kashani and Orvaschel (1988) reported that 17% of 14–16 year olds met the criteria for one or more anxiety disorders as defined in the DSM-III-R (APA, 1987) and 8.7% displayed symptoms severe enough to warrant treatment.

Although a limited number of fears and worries represent normal growth and development, concern arises when childhood fears and worries come so severe as to interfere with appropriate activities or have a negative impact on the child's relationships within the family, peers, or broader social networks. The implications of such difficulties may be lifelong, particularly if such difficulties prevent the child from learning appropriate social skills. Recent work suggests the possibility that children who suffer from anxiety mature into adults who continue to suffer from anxious symptoms (Last, Phillips, & Statfield, 1987). Last et al. (1987) report that many adults suffering with agoraphobia and generalized anxiety disorder describe symptoms of separation anxiety and generalized anxiety during their childhood years.

Although the question of effective treatments for more generalized anxiety has received little systematic research attention, behavioral techniques have been found to be effective in treating children's specific fears and phobias (King, Hamilton, & Ollendick, 1988). In the search for treatments of generalized childhood anxiety, investigators have drawn from the adult anxiety literature (see Barlow, 1988). As Strauss (1988) noted, generalized anxiety in adults has been successfully treated with various forms of behavior therapy. An effective treatment for generalized anxiety in children is hypothesized to be one that combines the demonstrated behavioral techniques with a focus on the cognitive components of anxiety. That there may be a cognitive component to childhood anxiety has been suggested recently (Kendall, Howard & Epps, 1988; Francis, 1988). Those studies that have examined children's cognitions as part of treatment for anxiety disorders have suggested that cognitions are important in both assessment and treatment (e.g., Morris & Kratochwill, 1983). The purpose of the present study is to evaluate the efficacy of a cognitive-behavioral treatment for diagnosed anxiety disordered children.

METHOD

Subjects

Four children, of at least average intelligence, with DSM-III-R diagnosed anxiety disorders completed the treatment protocol. Two children, a 10-year-old boy (JM) and a 13-year-old girl (DK), were diagnosed with Overanxious Disorder (Secondary Simple Phobia). The other two children were both 9-year-old girls diagnosed with Overanxious Disorder (CL, AM).

Two children were referred by school counselors after experiencing difficulties with anxiety (DK and CL) and two were identified on group screenings (JM and AM). Subjects were assessed using self-report measures, followed by individual structured diagnostic interviews. All subjects scored more than one standard deviation above the mean on both self-report measures and met DSM-III-R criteria for diagnosis.

Measures

Anxiety Disorders Interview Schedule for Children (Silverman & Nelles, 1988) is a structured interview schedule designed for the diagnosis of childhood anxiety disorders using both parent and child report. The schedule allows the assessor to screen out other disorders and is consistent with current DSM-III-R criterion.

The State-Trait Anxiety Inventory for Children (STAIC) (Spielberger, 1973) includes two 20-item self report scales: enduring tendencies to experience anxiety (A-Trait), and temporal and situational variations in levels of anxiety (A-State). Normative, reliability, and validity data are available (Morris & Kratochwill, 1983).

Child Behavior Checklist (CBCL) (Achenbach & Edelbrock, 1983) is an 118-item scale that assesses an array of behavioral problems and social competencies completed separately by parents and teachers.

State-Trait Anxiety Inventory for Children—Modification of Trait Version for Parents (A-Trait-P) (Strauss, 1987) is a modification of the trait version of Spielberger's (1973) STAIC to be used for parent report. To date, there are no reliability or validity data available for this measure.

Hamilton Anxiety Rating Scale, a 14-item independent clinician rating, was used to assess symptoms of anxiety (Hamilton, 1960). Slight modifications were made in the wording of the descriptors to accommodate children's language development.

Progress on target complaints (initial significant behavioral problems) was monitored throughout.

Design

Treatment was initiated after baseline assessment periods of 1, 2, or 3 sessions to produce a multiple-baseline design (Kendall & Koehler, 1985). Assessments were conducted at pretreatment, during treatment, posttreatment and again at follow-up.

Setting

The therapy was conducted through the Child and Adolescent Anxiety Disorders Clinic of Temple University. Some sessions were conducted at the child's school.

Intervention

Subjects received 16–20 one-hour, individual treatment sessions, meeting twice weekly except for illness and school holidays. The details of the treatment are specified in a treatment manual. The therapists were advanced graduate students who had participated in the development of the manual.

Children received training in the recognition and analysis of anxious cognitions and were taught to use them as cues for using anxiety management strategies to help them cope more effectively in anxiety-provoking situations. The treatment focuses on four major components: a) recognizing anxious feelings and somatic reactions to anxiety, b) clarifying cognitions in anxiety-provoking situations (i.e. unrealistic or negative attributions or expectations) c) developing

a plan to help cope with the situation (i.e. modifying anxious self-talk into coping self-talk as well as determining what coping actions might be effective), and d) evaluating the success of the coping strategies and self-reinforcement as appropriate.

Behavioral training strategies such as modeling, *in vivo* exposure, role play, relaxation training, and contingent reinforcement were incorporated. To help reinforce and generalize the skills, homework assignments were assigned. The training tasks and assignments were presented in a graduated sequence and were practiced in increasingly anxiety-provoking situations. Review of the previously introduced skills was included in each session. Once the full complement of skills has been acquired, these skills were practiced in imaginal and *in vivo* situations. Throughout the sessions, social reinforcement by the therapist was used to encourage and reward the child. In addition, the child was encouraged to verbally reinforce his own successful coping. The child was also urged to practice using the coping skills when anxiety provoking situations arose at home or at school.

Manipulation Check

An assessment of treatment integrity (Kazdin, 1986a) was completed via audiotapes. The second author listened to 20 randomly selected segments of randomly selected tapes (approximately 33% of the total number of tapes) and determined the integrity of the cognitive-behavioral procedures.

RESULTS

The effects of the program were positive. For all children, parent CBCL evidenced decreases from pre- to post-treatment on internalizing scales (See Table 1). Most gains were substantial enough to bring the internalizing behaviors to within normal limits (i.e., t score < 70). Both JM and DK continued to have some somatic problems, but at reduced levels. Parent reports of child anxiety were also obtained via weekly A-Trait-P reports. Inspection of Table 2 suggests that from pre- to post-treatment there was a decrease in the parent's ratings of each child's trait anxiety. All parents reported overall decreases in anxiety.

Independent clinician's ratings on the Hamilton indicated noticeable decreases in anxious symptoms (see Table 2). Based on the target complaints specific to each subject, the treatment produced decreases from at least moderate levels of anxiety to extremely mild levels. For example, CL initially had autonomic symptoms of anxiety such as sweating, tension headache, and dry mouth, which were reduced to very mild levels by the conclusion of treatment. Both JM and DK experienced gastrointestinal symptoms such as nausea in reaction to evaluation situations, which was substantially reduced by the treatment.

The children's A-Trait self-reports, using the multiple-baseline, evidenced change from pre- to post-treatment (see Figure 1). For each child, as the treatment was introduced across the multiple baseline, the reported levels of trait anxiety fell from the pretreatment level of at least one standard deviation above the mean, to a posttreatment level at or near the normative mean. As seen

TABLE 1
PARENTS CHILD BEHAVIOR CHECKLIST (CBCL) SCALES (T-SCORES)

Scales	Cases									
	Pre	JM Post	F-up	Pre	DK Post	F-up	Pre	CL Post	F-up	AM Post
1. Depression	82	71	71							
2. Depressed Withdrawal (Girls: 12-16)					68	69	74	55	64	60
3. Uncommunicative	74	65	70	85						
4. Obsessive/Compulsive (Boys: 6-11)	74	65	72							
5. Anxious/Obsessive (Girls: 12-16)				89	71	83				
6. Somatic Complaints	81	75	71	86	74	87	60	55	57	57
7. Social Withdrawal (Girls: 6-11)							62	56	55	59
8. Schizoid or Anxious (Boys: 6-11)										
9. Schizoid-Obsessive (Girls: 6-11)	72	61	71							
10. Schizoid (Girls: 12-16)				73	55	65	73	55	59	64
Total Internalizing	78	71	73	81	67	73	70	50	57	61

Note. Only internalizing scales are reported. Not all scales appear on all report forms. Blank spaces indicate scales that are not included.

TABLE 2
PARENT RATINGS (A-TRAIT-P) AND INDEPENDENT CLINICIAN RATINGS FOR
PRETREATMENT AND POSTTREATMENT ACROSS SUBJECTS

Subject	A-Trait-P Ratings		Clinician Ratings	
	Pre	Post	Pre	Post
JM	56	33	10	4
CL	48	32	10	3
DK	61	40	15	5
AM	52	35	9	1
Mean	54	35	11	3

Note. A-Trait-P = Parent ratings of child trait anxiety. Independent clinician ratings (using a modified Hamilton scale)

in Figure 1, levels of anxiety remained fairly constant during the baseline assessment periods except for AM, whose scores dropped briefly but returned to criterion prior to the onset of treatment. After treatment was instituted, the subject's reported levels of anxiety decreased, although the pattern of decrease differed among the children. JM reported a gradual decrease in trait anxiety levels over time, while both DK and AM reported fairly steady levels of anxiety until they had completed the first few *in vivo* exposure sessions. CL's reported levels of trait anxiety barely reached the criterion despite clear indications of anxiety on all other measures, and fell fairly quickly. Her reported levels of anxiety remained substantially below the mean for the remainder of treatment.

Target complaints for each child indicated behavior change. JM was initially unable to sleep in the dark and would become ill and go home when stressed at school; he now sleeps with only a nightlight, and does not leave school. His mother noted that JM was trying to help a fearful cousin by teaching him the elements of the treatment. DK initially experienced high levels of evaluation anxiety; at the end of treatment she was able to win a scholarship by performing in a competition with an audience. AM's grandmother had to pick her up from school everyday because of her fear, but AM became able to take public transportation or walk home from school alone. CL suffered significant evaluation anxiety, but, according to both parent and teacher reports, was able to concentrate and prepare for tests following treatment.

Teacher CBCL reports (see Table 3) for DK and CL indicated that pretreatment levels of internalizing behaviors were decreased to within normal ranges at posttreatment. Teachers' pretreatment reports for JM and AM indicated very low levels of internalizing and externalizing behaviors, and showed no change at posttreatment. Overall, teacher reports indicated lower levels of anxiety than other reports at both pretreatment and posttreatment. While the number of subjects in this sample is too small to draw any conclusion, additional research is needed to clarify these discrepancies across reporting sources.

Follow-up data indicated that for 50% of the children (AM, CL), according

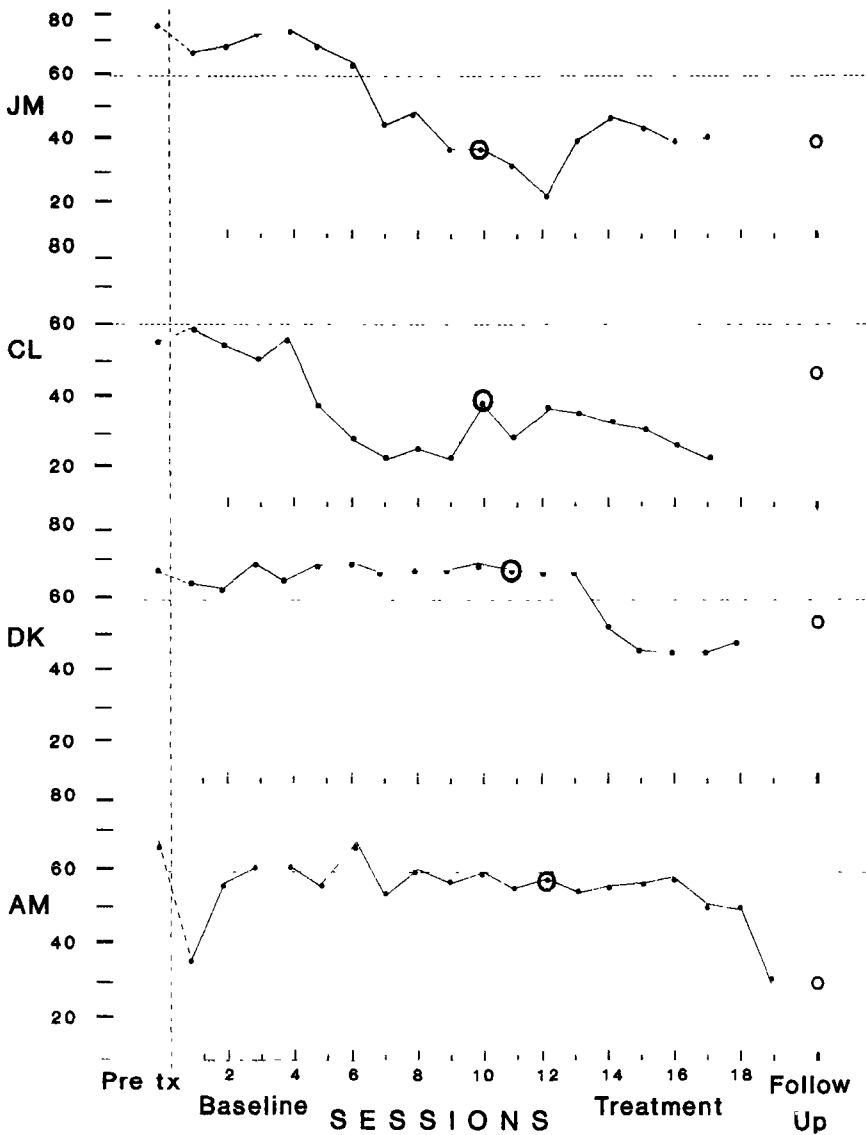


FIG 1. Changes in trait anxiety (T-scores) across treatment sessions for four cases. The circled data point indicates when the *in vivo* sessions began.

Note. The horizontal line at T-score 60 represents one standard deviation above the mean. Using this criterion for normative comparisons, the initial scores ($T > 60$) were reduced to within the normal range (1 S.D.).

TABLE 3
TEACHER CHILD BEHAVIOR CHECKLIST (CBCL) SCALES (T-SCORES)

Scales	Cases							
	JM		DK		CL		AM	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
1. Anxious	60	55	75	60	74	64	55	55
2. Social Withdrawal	60	55	63	55	55	55	55	55
3. Unpopular	55	55			55	55	55	55
4. Self-destructive	60	55	60	60	55	55	55	55
5. Obsessive/Compulsive (Boys: 6-11)	60	55						
6. Depressed			64	55	60	58	55	55
7. Immature (Girls: 12-16)			60	55				
Total Internalizing	60	55	68	60	65	61	55	55

Note. Only internalizing scales are reported. Not all scales appear on all report forms. Blank spaces indicate scales that are not included.

to both child and parent report, the anxious behaviors remained within normal ranges. For the other two children (DK, JM), the evidence of maintenance was qualified; the children's self-reports remained within normal ranges whereas parent's reports indicated some return of difficulty with anxious behaviors. JM's parent reported some internalizing problems, although not as problematic as at pretreatment. For DK, difficulties occurred on only those scales that had shown the least improvement at posttreatment.

DISCUSSION

Although preliminary, the results suggest that a cognitive-behavioral treatment can be efficacious in the treatment of anxiety disorders in children. Changes were produced on parent report, independent clinician's rating, and child self-report from pretreatment to posttreatment. Target behaviors were significantly improved. The use of a multiple baseline design helped to reduce the likelihood that the effects are simply due to sources of internal invalidity, such as the passage of time and contact with a therapist. However, the multiple-baseline design has limitations (see Kazdin, 1982) as does the use of only self-report assessments from multiple sources. A controlled group comparison outcome study using multimethod assessments is needed to investigate the efficacy of various strategies for the treatment of childhood anxiety disorders.

Follow-up data suggested that treatment gains were maintained for two of the children as reported on both child and parent report. The other two children reported continued gains on child self-report, but parent report evidenced a return of some previous difficulties. It is interesting that these two children had secondary diagnoses of Simple Phobia; the impact of multiple diagnoses on treatment outcome and maintenance of treatment gains is an important

question for future research. In addition, conversations with DK's parents revealed that she had experienced a number of unresolved medical problems and that her cousin had recently died of cancer. Such life stresses are anxiety provoking, particularly for a young person, and in part her increased difficulties may have been due to these situations. The potential benefits of booster sessions merits future attention.

A number of questions arose during treatment. The children ranged in age from 9 through 13, and the younger children seemed to have trouble identifying anxious cognitions in both imaginal and *in vivo* contexts, though they did report anxious ideation. They were able to discuss what might happen to another child in such a situation, but seemed unable to discuss their thoughts in terms of what might happen to them. Though tentative, these experiences suggest a developmental/age level below which anxious cognitions may be difficult to target. However, it is also possible that the inaccessibility of the self-talk was the result of a coping strategy being used by those particular children (C. Spielberger, personal communication, 1987).

Another issue concerns parental influence on the maintenance of anxious symptomology, and the implications for treatment strategies. Kendall, et al, (1988) noted that most behavioral patterns and cognitive interpretations have little or no meaning except within the context of social/interpersonal functioning. In each case the children expressed anxieties related to parental expectations and pressures or concerns about the welfare of their parent(s). Certain children found it difficult to alter their behavior patterns at home. Research is needed to study the role of family systems and specific parenting styles in the development and maintenance of anxiety disorders in children.

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